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Jérémie CHARRON, Vincent CAREY, Viviane MARCOTTE L'HEUREUX, Philippe ROY, Alain Steve COMTOIS, Pierre-Marc FERLAND

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Acute Effects of Cannabis Consumption on Exercise Performance – A Systematic and Umbrella Review

Jérémie CHARRON[1]*, Vincent CAREY[1], Viviane MARCOTTE L'HEUREUX[1], Philippe ROY[1], Alain-Steve COMTOIS[1] and Pierre-Marc FERLAND[1]

[1]Department of Exercise Science, University of Quebec in Montreal, Montreal, Canada

*Corresponding author: Alain Steve Comtois PhD, Department of Exercise Science, University of Quebec in Montreal, 141 Avenue du Président-Kennedy, H2X 1Y4, Montreal, Canada. E-mail: comtois.alain-steve@uqam.ca

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ABSTRACT

INTRODUCTION: The goal of this systematic and umbrella review was to regroup all systematic reviews, non-systematic reviews and all original articles into one convenient publication that would facilitate the theoretical and applied scientific investigations directed on cannabis consumption and exercise performance, to update current findings on the matters, and assess evidence quality.

EVIDENCE ACQUISITION: The systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) method. A computer-based systematic search was conducted in September 2019 through the Pubmed, Scopus and SPORTDiscus databases. The reliability of the systematic search was assured by having the article selection process entirely repeated by a second author. Strength of evidence of the selected articles was assesses using a modified version of the Downs & Black Checklist.

EVIDENCE SYNTHESIS: The systematic search yielded a total of 8 peer-reviewed publications as well as 10 literature reviews. Results show that cannabis consumption prior to exercise induces decrements in performance (reduced ability to maintain effort, physical/maximal work capacity), undesired physiological responses (increased heart and breathing rate as well as myocardial oxygen demand) and neurological effects on balance (increased sway).

CONCLUSIONS: Based on the articles included in this review, the authors conclude that cannabis consumption has an ergolytic effect on exercise performance and therefore does not act as a sport performance enhancing agent as raised by popular beliefs. Thus, cannabis consumption prior to exercise should be avoided in order to maximize performance in sports. Further research should mimic modern THC dosage (150 mg).

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Key words: marijuana, physiology, ergolytic, tetrahydrocannabinol, THC, sports, cardiovascular

(INSERT GRAPHICAL ABSTRACT HERE)

TEXT

INTRODUCTION

Cannabis is a plant that contains more than 400 compounds, including 60 cannabinoids [1]. The two most commonly known compounds are tetrahydrocannabinol (THC) and cannabidiol (CBD) [2]. THC is more psychoactive than the latter as CBD acts as an additive and synergistic or antagonist to THC [1]. THC is largely responsible for the psychological and physical effect it has on someone as opposed to CBD which is not psychotropic and can, in fact, act as an anticonvulsant and attenuate psychological effects caused by THC [1].

Researchers believe that the effect of cannabis is modulated by the nervous system and two cannabinoid receptors (CB1 and CB2) [3]. The CB1 receptors contribute to the psychotropic effects of cannabis and are located in the brain and in the peripheral nerves [4, 5] while the CB2 receptors are located in macrophages sensory tissues of the spleen as well as in other immune cells [6]. The two most common ways to consume cannabis is through smoke inhalation, which cause an effect within minutes, or edible ingestion, which will delay the effect by 30 minutes up to two hours [1]. Smoke inhalation will result into a quicker intoxication because THC will be absorbed through the lungs and quickly released into the bloodstream where it will quickly reach the brain. In contrast, edibles must be ingested first via the gut before being metabolized by the liver. This will stretch the cannabis' effect duration as 65% of its metabolites will go back into the intestine and be reabsorbed [1]. Cannabis is an illegal drug in most countries and was added to the International Olympic Committee (IOC) banned substance list in 1989[7].

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Despite being an illegal drug and banned substance in sport setting, the popularity of cannabis has significantly increased since the early 2000's, with its consumption significantly increasing from 17.3% to 19.6% between 2002 and 2014 among adults aged between 18-25 years old living in the USA [8]. While some countries struggle with the consumption of illegal cannabis [9], parts of the US and other countries have legalized it for recreational use [9] consequently making cannabis easier to access [10]. In 2018, Canada was the third country to follow this movement, three years after Uruguay [9]. As of October 2019, Georgia (the country) and South Africa have also legalized the recreational use of cannabis [11, 12]. In the US, there is currently only 5 states that have legalized recreational cannabis: Alaska, Colorado, Oregon, Washington and District of Columbia; meanwhile several others (21 states) have decriminalized its possession [10].

At the time this study was conducted, the literature presented four main reasons for cannabis consumption: its euphoric effect, the fact that it decreases anxiety and helps to relax, its social aspect, and an alleged sports performance-enhancing effect [1, 7, 13, 14]. Furthermore, a recently published systematic review concluded that Caucasians were more prone to cannabis use in order to enhance sport and recreational performance [15]. Additionally, a study by Lorente *et al.* conducted on French sport students showed that 24.2% of cannabis consumers use it to seek pleasure (euphoric effect), 36.1% to reduce stress and 20.9% to increase pleasure during parties because of the increase in sociability. Their study also highlights several sports where athletes are more prone to cannabis consumption, interestingly, all are considered sliding sports that include skiing and kite surfing, snowboarding, skeleton, bobsleigh, hockey, and skateboarding. Athletes from these sports would consume cannabis during competition seeking for similar sensations as when they leisurely practice their respective sport (exhilaration), and would increase cannabis consumption as their competition level increased [14]. Also, two reviews by Pesta *et al.* and

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Saugy *et al.* highlighted that some athletes use cannabis to reduce pre-competitive stress [4, 16]. In addition, another study stated that adolescent athletes were less prone to consume cannabis than non-athletes of their age [17]. A recent study that observed performance markers on a control group and chronic cannabis physically active consumers showed that chronic consumption of cannabis does not seem to negatively affect performance [18]. Another study distributed a survey and showed that cannabinoids represent 48.5% of performance enhancing drugs (PEDS) used by ultramarathoners [19]. Authors suggested that runners may use cannabinoids because of supposedly improved vasodilation and bronchodilation that would help with oxygenation, therefore helping performance[19].

At the moment, 3 systematic reviews and 7 reviews have been published on cannabis consumption and performance. Nonetheless, the most recent systematic review was published in 2017 by Kennedy [20], which was submitted for publication on August 25th 2016. This means that almost 4 years of potential new studies in online databases could update current knowledge on a present-day concern related to national and international sporting bodies, since several governments have now legalized cannabis.

Thus, the goal of this systematic and umbrella review was to regroup all systematic reviews, reviews and all original articles into one convenient publication. This would facilitate the theoretical and applied investigations directed on cannabis consumption and exercise performance, update current findings on the matters, assess evidence quality, provide assistance in decision making for governing bodies, and offer concise tables that could be used for communication to a general public so as to increase knowledge and prevent drug misuse in athletes.

Furthermore, results could help professionals better understand the acute effects of cannabis consumption on exercise performance and help them increase awareness towards their

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athletes. In addition, results could also highlight a deficiency in the current literature and help guide future research projects.

EVIDENCE ACQUISITION

The systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) method. The databases utilized in this study were three of the four databases suggested by the University of Québec in Montréal (UQÀM) library services: Pubmed, Scopus and SPORTDiscus. ERIC (ProQuest) was discarded due to its educational focus. The systematic search started in September 2019 using the following search strategy in titles, abstracts and keywords: (cannabis OR Cannabi* OR marijuana OR marihuana OR THC OR tetrahydrocannabinol OR «Delta-9-tetrahydrocannabinol») AND (exercise OR Effort OR «physical exertion» OR «maximal exercise» OR «Submaximal exercise») AND («sportive performance» OR performance). All search results were extracted into EndNote X9.

Inclusion and Exclusion Criteria

Duplicates

After placing all article titles in alphabetical order in EndNote X9, the duplicates were manually discarded. If an article title was presented twice but with a different year of publication or a different journal, it was considered as different and was not discarded.

Titles & Abstracts

Subsequently, if the authors could clearly identify by reading the title or the abstracts that the article did not present an evaluation of the effect of cannabis consumption on human exercise performance, the article was discarded. In the beginning, only original investigations were considered. Later, the authors decided to improve this review by adding the umbrella review in order to present a more complete representation of the subject after encountering multiple interesting reviews.

Language

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If the text was not written in English, the article was discarded.

Text

Finally, when screening the text, the authors first confirmed that the subjects were humans and then screened for performance, physiological or balance assessment measurements in direct response to exercise or balance tasks after acute cannabis consumption. Acute consumption was defined as either having smoked cannabis in the last four hours before testing and in the last 12h for cannabis ingestion as clinical effects of cannabis can last this long [21]. Studies on long term effects associated with chronic consumption of cannabis were not included as this heterogeneity in study design would make comparison difficult.

Manual Search

A manual search was performed through the reference list of all articles retained from the systematic search as well as systematic and literature reviews from the full-text criteria. To be included in the umbrella review, reviews had to address relationship between cannabis consumption and exercise performance.

Procedures

Extraction

All results were extracted to an Excel sheet.

Verification Methods

To ensure the systematic search reliability, the article selection process was entirely repeated by a second author, which also validated the Modified Downs & Black. If there was any disagreement, a third author took the final decision.

Modified Downs & Black

All selected articles of the systematic review were then evaluated using a modified version of the Downs & Black checklist [22] to assess strength of evidence. Our version used the modification on the 27th item as suggested by Korakakis *et al.* [23] (which is question 26 in our version) and added the three following items; #10 Were the results significant at <0.05? #11

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Were the results significant at <0.01? #16 in order to increase statistical significance. Was the attempt to blind measures successful? Studies included in this review usually mentioned in their article whether the blinding was successful or not. These items were added because it further validated how statistically significant the study results were. Furthermore, the following items from the original Downs & Black Checklist were removed because they did not apply to the current theoretical research; *Have the characteristics of patients lost to follow-up been described?, Were losses of patients to follow-up taken into account?*. Both theses items were removed because the current study observed acute responses and therefore the studies included had no follow-ups. Also, the following items were removed because the included studies did not analyse effect of a treatment; *Were the staff, places, and facilities where the patients were treated, representative of the treatment the majority of patients receive?, Was there adequate adjustment for confounding in the analyses from which the main findings were drawn?*. Following the removal and addition of questions, ratings associated with poor, fair, good and excellent evidence were recalculated. Ratings between 24-27 were considered Excellent, 19-23 Good, 13-18 Fair, and less than 13 was considered Poor evidence.

Meta-Analysis

Common measures across the studies such as heart rate (HR), systolic blood pressure (SBP) and diastolic blood pressure (DBP) were regrouped to perform random effects inverse variance forest plots with confidence interval (CI) set at 95%. The analysis was performed on RevMan 5.4.

EVIDENCE ANALYSIS

The systematic and manual search yielded a total of 8 peer-reviewed publications containing relevant data [24-31]. The umbrella review resulted in 3 systematic reviews [15, 20, 32] and 7 non-systematic reviews [2, 4, 7, 16, 33-35]. A total of 18 publications were included in this review (See

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Figure 1.) [2, 4, 7, 15, 16, 20, 24-35]. Tables 1 and 2 present results from the systematic and umbrella reviews. Throughout this search, we have encountered some interesting studies that unfortunately did not meet our inclusion criteria, such as observation of cannabinoid release during exercise on abstinent users, effect of progressively increasing daily cannabis dosage over a 28 day period and comparing heavy users with occasional users after an abstinence period on neuropsychological assessments, resting measures or exercise.

(INSERT FIGURE 1 HERE)

Results show that cannabis consumption prior to exercise induces decrements in performance (reduced ability to maintain effort, physical/maximal work capacity), undesired physiological responses (increased heart rate, breathing rate and myocardial oxygen demand) and neurological effects on balance (increased sway) (view table 1). Authors included in the umbrella review conclude that cannabis consumption prior to exercise does not enhance sports performance (view table 2). Thus, one could conclude that cannabis acts as an ergolytic and that cannabis should not be consumed in order to improve sports performance.

The strength of evidence analysis based on the Modified Downs & Black Checklist shows that most of the included studies in the systematic review have good or fair evidence. No study received the mention of «poor evidence» and one received an excellent rating.

(INSERT TABLE 1 HERE)

Systematic Review

Physiological Response

The results presented in Table 1. show that cannabis consumption significantly increases heart rate (up to 51% during recovery and 18% during exercise) [25]. Research has shown that tachycardia persisted up to 80% of maximal effort but was not significantly present at 90% and 100% [28]. Not all research presented the same quantitative results regarding tachycardia, these differences could be explained by the fact that tachycardia induced by cannabis consumption is dose-related [36] and that the studies included in this research utilized different

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THC dosage in their protocols, ranging from 7.5 mg to 50.6 mg. This could be explained by the fact that the majority of those studies were published between 1974 and 1986 and that the THC dosage was estimated between 5 and 25 mg per joint during that time period whereas in today's joints, the THC concentration may be as elevated as 150 mg per joint [1]. Since the effects of cannabis consumption are dose-related, authors suggest that further research design include protocols with higher dosages, which would recreate more accurately modern day cannabis consumption.

Another factor that could explain disparity between the studies results is timing of consumption and exercise. In fact, the studies presented various delays between cannabis consumption and exercise, such as 20, 40, 90, 100 and 210 minutes have been employed [25, 27-29, 31]. Furthermore, both studies published by Aronow & Cassidy did not clearly mention the delay between cannabis consumption and exercise. We estimated the delay to 10-15 minutes after examination of their methodology [24, 30]. The delay between cannabis consumption and exercise will affect research results as the peak cannabis effect happen 20-30 minutes after its consumption when smoked [37] and 3 hours when ingested [38]. Despite the peak occurring later as mentioned above, both studies from Aronow & Cassidy [24, 30] used a 10 to 15-minute delay between cannabis inhalation and exercise testing in their protocols. These studies' measurements might have been taken too early to examine the effect of cannabis consumption on exercise performance during peak intoxication. In contrast, the studies published by Avakian *et al.*, Steadward & Singh and Bird *et al.* whose protocols waited 40 [25, 29] and 100 minutes, [31] respectively, before exercise testing might of waited too long and therefore did not test during peak intoxication. On the other hand, the study published by Newmeyer *et al.* presented a protocol with 90 and 210 minutes [27] of delay between cannabis consumption through edibles and exercise testing, authors concluded that these responses were measured during peak intoxication.

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The data collected from blood pressure (BP) measurements in response to cannabis consumption is even more mitigated than the one collected from tachycardia. In fact, three studies presented an increase in BP [24, 29, 30], two studies presented a decrease in BP [26, 28] and one study presented no significant effect on BP [25]. The three studies presenting an increase in BP were conducted on cigarette smokers [24, 30] and marijuana smokers [29], which is surprising since a previous study showed that tobacco smoking does increase BP but that cigarette smokers also have lower BP than healthy non-cigarette smokers [39]. Furthermore, three studies observed an increase in the product of HR and systolic blood pressure (SBP) which is an indicator of myocardial oxygen consumption [24, 25, 30]. The increase in the product of HR & SBP could be explained by the increase in carboxyhaemoglobin (COHb) as observed by Aronow and Cassidy in their two publications [24, 30]. An increase in COHb limits the amount of hemoglobin oxygen saturation that can be delivered to the heart, consequently causing an oxygen delivery lower than the oxygen demand.

The effect on the respiratory system was presented as breathing rate that was measured in one study by Renaud & Cormier [28]. The results indicated a greater than normal increase in breathing rate up to 80% of maximal effort, similar to what was observed with HR [28].

Overall, the previously mentioned factors (increased HR, BP, breathing rate, myocardial oxygen consumption and COHb) could explain the drop in performance as expressed by decreased in exercise time, maximal work capacity (MWC) and physical work capacity (PWC) observed by several authors [24, 28-30].

Performance measures

In cigarette smokers that consume 20 cigarettes/day, the drop in exercise time reached 50% [24], whereas in healthy young participants the MWC dropped by 6.2% [28] and PWC dropped from 1177.2±65.4 to 829.9±50.3 PWC_{170}Kgm/min [29]. This difference could potentially be explained by the fact that cigarette smokers tend towards a more sedentary lifestyle and have

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higher perceived exertion rating than non-cigarette smokers [40]. There are no similar results with cannabis consumption, however, one early study shows that performance in the handgrip strength test is not affected by cannabis consumption [29].

Further research could be directed towards comparing handgrip strength of chronic cannabis smokers to a healthy control group as a study by Al-Obaidi *et al.* previously showed that cigarette smokers have reduced grip strength and fatigue faster than healthy non-cigarette smokers [41].

Neurological Response

The one leg stand (OLS) and walk and turn test (WAT) were significantly affected by oral cannabis consumption as shown in Newmeyer *et al.*'s study as the number of clues increased (such as losing balance, not walking on the line, using arms for balance, etc.) by 1.99 and 1.6 respectively. The modified Romberg balance test (MRB) was also significantly affected as an increase in sway was observed [27]. An increase in sway was also observed in another study that noted significant impairments on standing steadiness [31]. This increase might be due to dizziness, a known effect of cannabis consumption [7].

Ultimately, the cardiovascular, pulmonary and neurological systems are all affected by cannabis consumption. Consequently, cannabis can be considered an ergolytic and should be avoided in order to maximize sports performance. Overall, the pool of studies included in this systematic review is very heterogeneous and differences in participants should be taken into account when interpreting the data. Also, it should be noted that studies included in this review were of fair to good evidence based on a modified version of the Downs & Black Checklist.

(INSERT FIGURE 2 HERE)

Strength of Evidence Assessment

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Each study's results on the modified Downs & Black were regrouped to form an overall score for each category (See Figure 2.). Reporting was worth 96 points (8 studies x 10 1-point question and one 2-points question). This pool of studies obtains a high score of 83/96 (86.5%). External validity was worth 16 points, and the overall score was 0 point. Internal Validity – Bias was worth 72 points and the overall score was 63/72 (87.5%). Internal Validity – Selection Bias was worth 24 points and the overall score was 21/24 (87.5%). Finally, Power was worth 8 points and the overall score was 1/8 (12.5%). The lowest scores were obtained on both categories that weight the less in the balance with only one question for Power and two questions for External Validity. Therefore, even thought two categories failed, this pool of studies can be considered as robust and of great overall quality.

(INSERT FIGURE 3 HERE)

Meta-Analysis

Results from the meta-analysis shows that HR, SBP and DBP are almost always higher after cannabis consumption. Usually, a meta-analysis should not be performed on heterogenous studies[42], however, in this case, even thought the studies are very heterogenous (I^2 = 99 %, 79% and 97% for HR, SBP and DBP respectively with p=0.0001), the confounding factors that made those studies so disparate (such as delay of consumption, THC dosage and participant characteristics) do not seem to affect the end results which is a net increase in all values (See Figure 3.). The mean difference was -26.88 with a CI at 95% of [-35.69, -18.07] for HR, -9.51 [-12.08, -6.95] for SBP and -6.86 [-12.91, -0.81] for DBP.

(INSERT TABLE 2 HERE)

Umbrella Review

The systematic research yielded a total of 10 reviews presenting relationships between cannabis consumption and sports performance [2, 4, 7, 15, 16, 20, 32-35]. Out of those 10 reviews, 4 concluded that cannabis consumption does not enhance performance and is in fact an ergolytic agent [4, 7, 20, 33]. The first systematic review, published in 2017 by Kennedy [20],

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included several studies that were not specifically related to sport performance or the cannabis was taken post-exercise or were not peer-reviewed original articles (some were abstracts). Therefore, this review presented a wide array of articles on outcome factors that were affected by THC. The methodology was also very brief, making it hard to know how rigorous the research was. Additionally, Eichner based his conclusion on 3 studies [28, 31, 43], two of which were included in our review [33] and the last one was an abstract. Meanwhile, Campos *et al.* based their conclusion only on Eichner's review [7] which is basically a review reviewing a single review. The conclusions of both of these reviews were based on very limited data. Finally, Pesta *et al.* based their results on 9 articles [14, 25, 28, 29, 31, 43-46], with some being a case report on a 13-year-old boy who experienced bronchodilatation **after** smoking cannabis in a study that stimulated a bronchospasm by exercise and then made participants smoke cannabis to observe the decrease of this bronchospasm, another study where spirometry measurements were taken without exercise, another study directed on the prevalence and reasons behind cannabis consumption in French University students and finally, the latter study was an abstract.

The systematic review by Trinh *et al.* in 2018 which included 3 articles [28, 29, 47]concluded that the effect of cannabis remained unclear due to the low number of quality studies available This review is the only one so far that actually evaluated the included articles (GRADE assessment). Likewise, Ware *et al.* did not officially present a negative effect of cannabis consumption on exercise performance, their conclusion stated that there was no direct evidence of performance-enhancing effects of cannabis in athletes [2]. Their investigations' conclusion was based on only 2 articles [28, 29] Ultimately, the previously mentioned reviews all tend towards highlighting the harmful effects of cannabis on performance even though most conclusions were based on limited data.

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Another review, by Saugy *et al.* suggests taking measures to avoid players misusing cannabis as it can be detected up to 5 days post-consumption and could result in positive doping test [16]. Saugy *et al.* presented a mixed opinion on the subject, mentioning the negative effect on performance by citing Renaud & Cormier's research [28] but also mentioning that cannabis could potentially help athletes relax and avoid social pressure. After all, only one article [28] included in their review presented a direct effect of cannabis on exercise performance.

Hilderbrand's review concluded that athlete's use of cannabis prior to competition may lead to impairment or inappropriate decision-making and present a danger for themselves or others [34]. His conclusion was based on several articles on driving performance and not from research directed on actual athletes, making his conclusion not evidence based.

Finally, the last review included in this research is the systematic review conducted by Brisola-Santos *et al.* in 2016 [15] which concluded that some subgroups such as male, Caucasian and sliding sports athletes seemed to use it to enhance performance. This further justifies researching on the subject to increase awareness and avoid cannabis misusing.

Finally, most reviews agree that cannabis does not act as a sport-enhancing substance and that it is, in fact, an ergolytic agent. This review highlighted that most of the reviews presenting the effect of cannabis consumption on exercise performance based their conclusions on very limited data, were not systematic (except 3) and did not assess evidence quality (except Trinh *et al.*[32]). The only very thorough and detailed review with a well-developed methodology, bias assessment and evidence quality assessment was by Trinh *et al.* Unfortunately, this review only included three studies.

Conclusion

Based on the articles included in the current review, the authors conclude that cannabis consumption has an ergolytic effect on exercise performance as it increases HR, affects

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balance, decreases ability to maintain effort, increases myocardial oxygen demands and decreases physical work capacity. These results should be interpreted carefully as the samples presented in the cited studies were very heterogeneous. Meanwhile, most of the reviews included in this research conclude that cannabis does not enhance exercise performance, but were based on very limited data or on a methodology that was either non-systematic or not well detailed (except one).

In addition, the results presented herein could help professionals better understand the acute effects of cannabis consumption on exercise performance and help them increase awareness towards their athletes. The results also highlight a lack in the current literature and could help guide future research projects.

Moreover, this review, to the best of our knowledge, is the first to regroup all previous reviews and critically assessed the robustness of their conclusions. We trust that the information presented herein will help professionals better understand the risks associated with cannabis consumption, and the graphical abstract and simple tables will facilitate communication of this grouped information to a general public and to athletes that will hopefully help unravel the misbeliefs that cannabis can enhance performance. Further research should include modern THC dosage (150 mg), be directed on the use of cannabis and stress relief in athletes, and report the chronic effects of cannabis consumption on exercise performance.

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NOTES

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Authors contribution:

Jérémie Charron: Methods, Results, Discussion, Conclusion, Editing and reviewing
Viviane Marcotte L'Heureux: Methods
Vincent Carey: Results and Tables
Philippe Roy: Introduction
Alain Steve Comtois: Methods, Editing and reviewing
Pierre-Marc Ferland: Introduction, Methods, Editing and reviewing
All authors read and approved the final version of the manuscript.

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Table 1. Summary of the Reviewed Articles Presenting Responses to Exercise Following Cannabis Consumption

Authors	n	Age	Delay (min)*	Consumption method	Participants	THC dosage	Physical test	Response	Modified D&B Score	Evidence Rating
Bird *et al.*[31]	161	18-36	100	Oral	Healthy university students volunteers	10-20mg	Standing steadiness Complex reaction time VDA Pursuit-rotor	↓ standing steadiness ↓psychomotor coordination ↓Reaction speed ↑ pulse rate Conjunctival Hyperaemia	23	Good
Steadward & Singh[29]	18	21-27	40	Inhaled	MJ smokers	18.2mg	Modified submaximal Sjostrand bicycle ergometer test Handgrip strength	↑ HR ↓ BP ↓ PWC No effect on handgrip strength No effect on vital capacity or expiratory flow	21	Good
Newmeyer *et al.*[27]	20	18-50	90 & 210	Inhaled, vaporized and oral	Good physical health, occasional & frequent smokers	50.6mg	MRB OLS WAT	Minimal impairment on MRB ↑number of clues observed for OLS & WAT	25	Excellent
Aronow & Cassidy (1974)[30]	10	47.3 (± 6.1)	Not specified	Inhaled	Patient with angina, smoking 20 cigarettes/day	19.8mg	Progressive workload on ergocycle until onset angina. 25W for 180s., 50W for 150s., 75W until onset angina	↓by 48% on exercise time until onset angina	23	Good
Aronow & Cassidy (1975)[24]	10	49.6 (± 5.8)	Not specified	Inhaled	Patient with angina, smoking 20 cigarettes/day	18.9mg	Progressive workload on ergocycle until onset angina; 25W for 180s., 50W until onset angina	↓by 50% exercise time before onset angina ↑ HR ↑ Myocardial oxygen demand ↑COHb level	22	Good
Renaud & Cormier[28]	12	20-24	20	Inhaled	Healthy young participants	1.7% of THC in a 7mg/kg cannabis joint	Progressive exercise testing on ergocycle until exhaustion	↓ MWC ↑breathing rate Slight increase in VO_2 and VCO_2 at high workload No detectable COHb increase	17	Fair
Avakian *et al.*[25]	6	21-27	40	Inhaled	Chronic MJ users with no involvement in athletic programs	7.5mg	15 min on ergometer bicycle at 50% V02max	↑ HR during exercise and recovery No significant effect on all other variables	19	Good
Hollister *et al.*[26]	17	21-44	Not specified	Oral	Participants in good general and mental health, not currently taking any drugs, and not undergoing psychotherapy	30-70mg Median of 50mg	Muscle strength measured by finger ergograph	Marked dizziness. Weakness was clearly demonstrable on the finger ergograph ↓BP	18	Fair

Oxygen uptake **VO_2**; Carbon dioxide production **VCO_2**; Carboxyhemoglobin **COHb**; Modified **D&B**; Brown & Black

**delay between cannabis consumption and physical testing*

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Table 2. Summary of the Sources and the Conclusions from the included Reviews

Authors (year)	Title	Articles included	Conclusion
Brisola-Santos *et al.* (2016)[15]	Prevalence and correlates of cannabis use among athletes-A systematic review	Ama *et al.*, 2002; Buckman *et al.*, 2009; Buckman *et al.*, 2013; Buckman *et al.*, 2011; Diehl *et al.*, 2014; Dunn *et al.*, 2011; Green *et al.*, 2001; Laure *et al.*, 2004; Laure *et al.*, 2007; Lorente *et al.*, 2005; Maquirriain, 2010; Peretti-Watel *et al.*, 2003; Peretti-Watel *et al.*, 2004; Thevis *et al.*, 2008; Van Eenoo & Dalbeke, 2003	This review suggests that a number of athletic subgroups are at increased risk for marijuana use. A common rationale for use appears to be to enhance sports performance
Kennedy (2017)[20]	Cannabis: Exercise performance and sport. A systematic review	Aronow & Cassidy, 1974; Aronow & Cassidy, 1975; Avakian *et al.*, 1979; Benowitz *et al.*, 1975; Renaud & Cormier, 1986; Shapiro *et al.*, 1976; Steadward & Singh, 1975; Tashkin *et al.*, 1975; Westin *et al.*, 2014; Wong *et al.*, 2013; Hollister *et al.*, 1968	THC does not enhance aerobic exercise or strength
Trinh *et al.* (2018)[32]	Marijuana and Its Effects on Athletic Performance: A Systematic Review	Renaud & Cormier, 1986; Steadward & Singh, 1975; Maksud & Baron, 1980	Because the number and quality of studies was low, the effects of marijuana on athletic performance remain unclear
Ware *et al.* (2018)[2]	Cannabis and the Health and performance of Elite Athlete	Kennedy 2017; Brisola-Santos *et al.,* 2016; Steadward 1975; Renaud 1986; Eichner 1993; Lorente 2005; Saugy 2006; Catlin *et al.,* 1996; Campos *et al.,* 2003; Hilderbrand 2011	Although cannabis use is more prevalent in some athletes engaged in high-risk sports, there is no direct evidence of performance-enhancing effects in athletes
Pesta *et al.* (2013)[4]	The effects of caffeine, nicotine, ethanol, and tetrahydrocannabinol on exercise performance	Steadward & Singh, 1975; Renaud & Cormier, 1986; Avakian *et al.*, 1979; Tashkin *et al.*, 1977	Only caffeine has enough strength of evidence to be considered an ergogenic aid. Cannabis and alcohol are ergolytic for sports performance, and nicotine needs confirmation with further research.
Hilderbrand (2011)[34]	High-performance sport, marijuana, and cannabimimetics	Huestis 2002; Pipe 2009; Hindrik & O'Hanlon, 1993; Smiley, 1986; Gerberich *et al.*, 2003; Mura *et al.,* 2003; Ramaekers *et al.,* 2004; Moving *et al.*, 2004; Drummer *et al.*, 2004; Laumon *et al.,* 2005; Ronen *et al.,* 2008; Karschner *et al.*, 2009; Canfield *et al.,* 2010	The use of marijuana by the elite athlete prior to competition may result in danger to that particular athlete or others as a result of impairment of response or inappropriate decision making.
Huestis *et al.* (2011)[35]	Cannabis in Sport Anti-Doping Perspective	Steadward & Singh, 1975; Renaud *et al.,* 1986; Saugy *et al.,* 2006; Wagner *et al.,* 1986; Catlin *et al.,* 1996; Red *et al.,* 2003	Recent advances in understanding the endogenous cannabinoid system demonstrate its important role in many critical functions that could positively affect sports performance.
Saugy *et al.* (2006)[16]	Cannabis and sports	Bird *et al.*, 1980; Lach & Schachter, 1979; Tashkin *et al.*, 1975; Vachon *et al.*, 1973; Lorente *et al.*, 2005; Renaud *et al.,* 1996; Arvers *et al.,* 2003	There is an urgent need for sport authorities to take measures necessary to avoid players misusing cannabis
Campos *et al.* (2003)[7]	Marijuana as doping in sports	Eichner, 1993	The intent to use them to enhance performance will fail
Eichner (1993)[33]	Ergolytic drugs in medicine and sports	Tashkin *et al.*, 1977; Bird & al, 1980; Renaud & Cormier, 1986	Marijuana makes the heart work harder and curtails exercise performance. Marijuana is therefore considered as an ergolytic drug

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Titles of Figures

Figure 1. Flow diagram of the search strategy and article selection process

Figure 2. Modified Downs & Black overall scores

Figure 3. A) Forest Plot of the effect of cannabis on heart rate B) Forest Plot of the effect of cannabis on systolic blood pressure C) Forest Plot of the effect of cannabis on diastolic blood pressure

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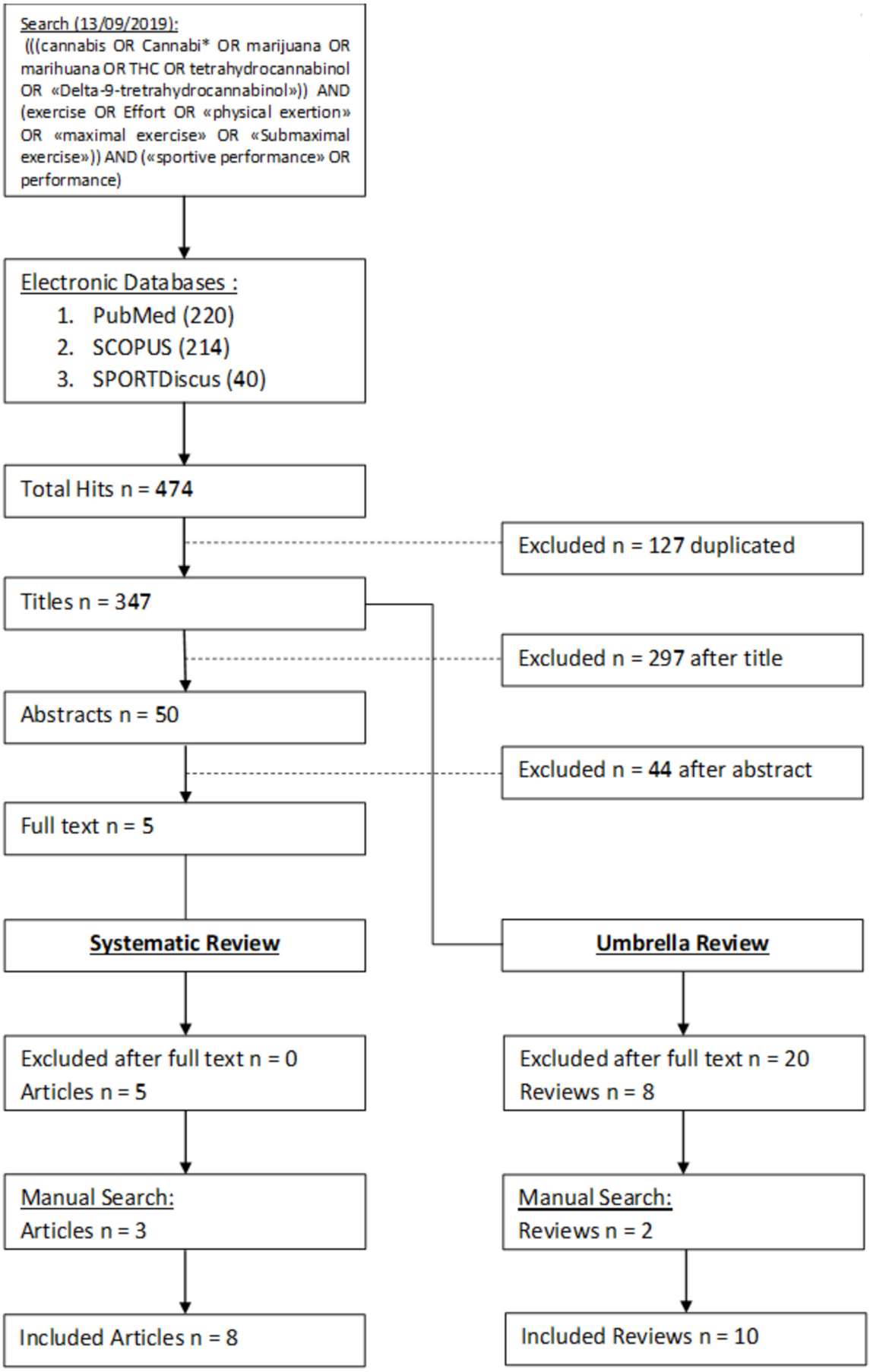

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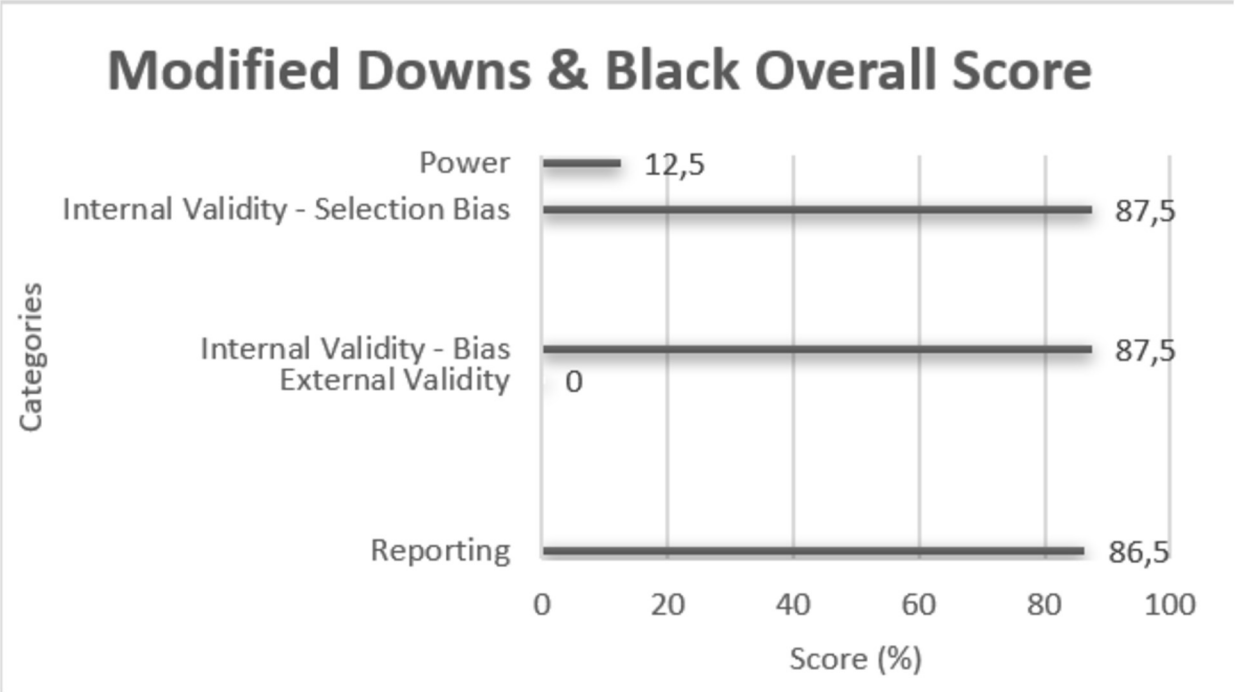

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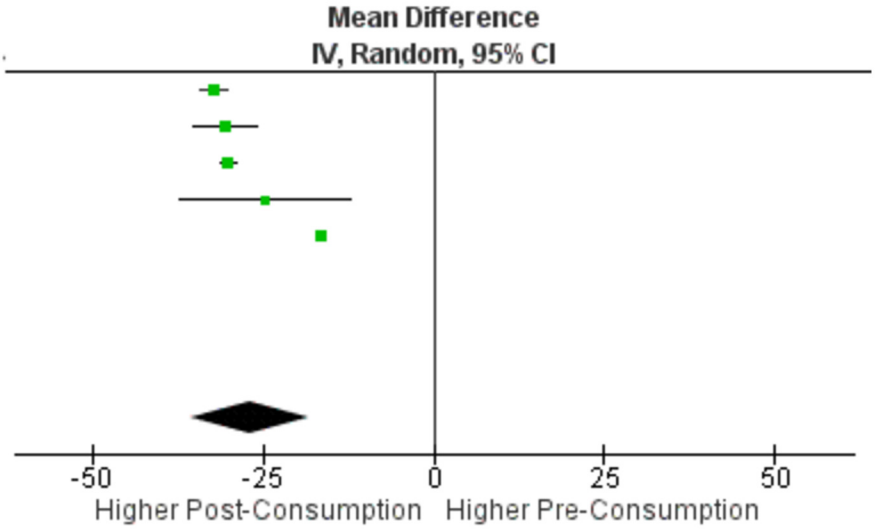

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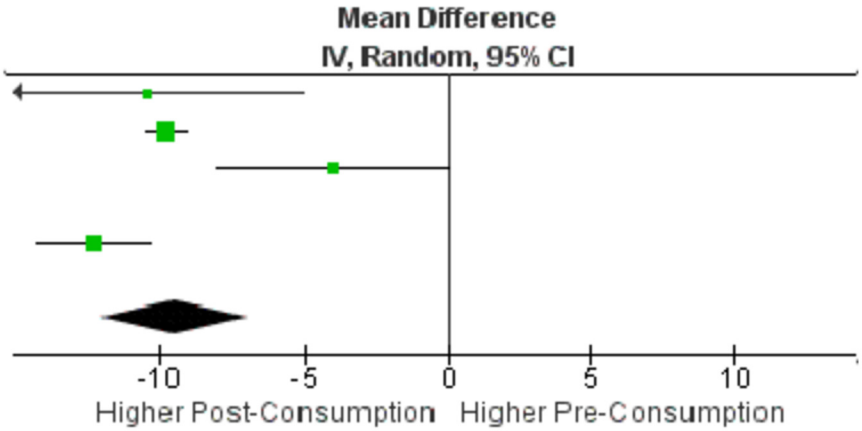

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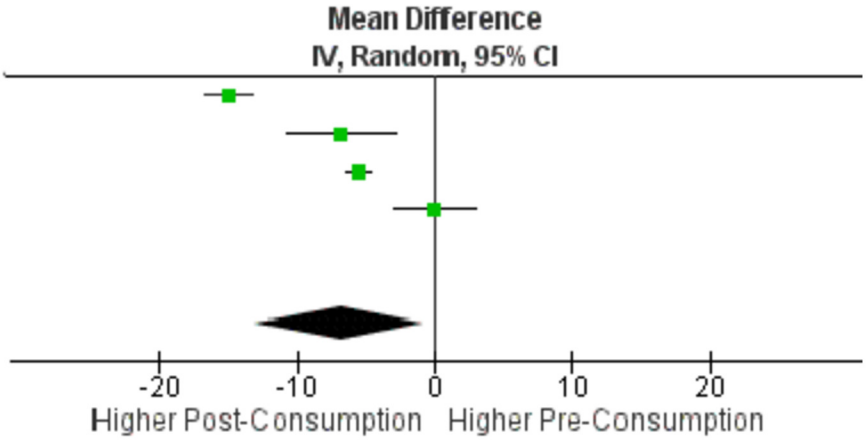

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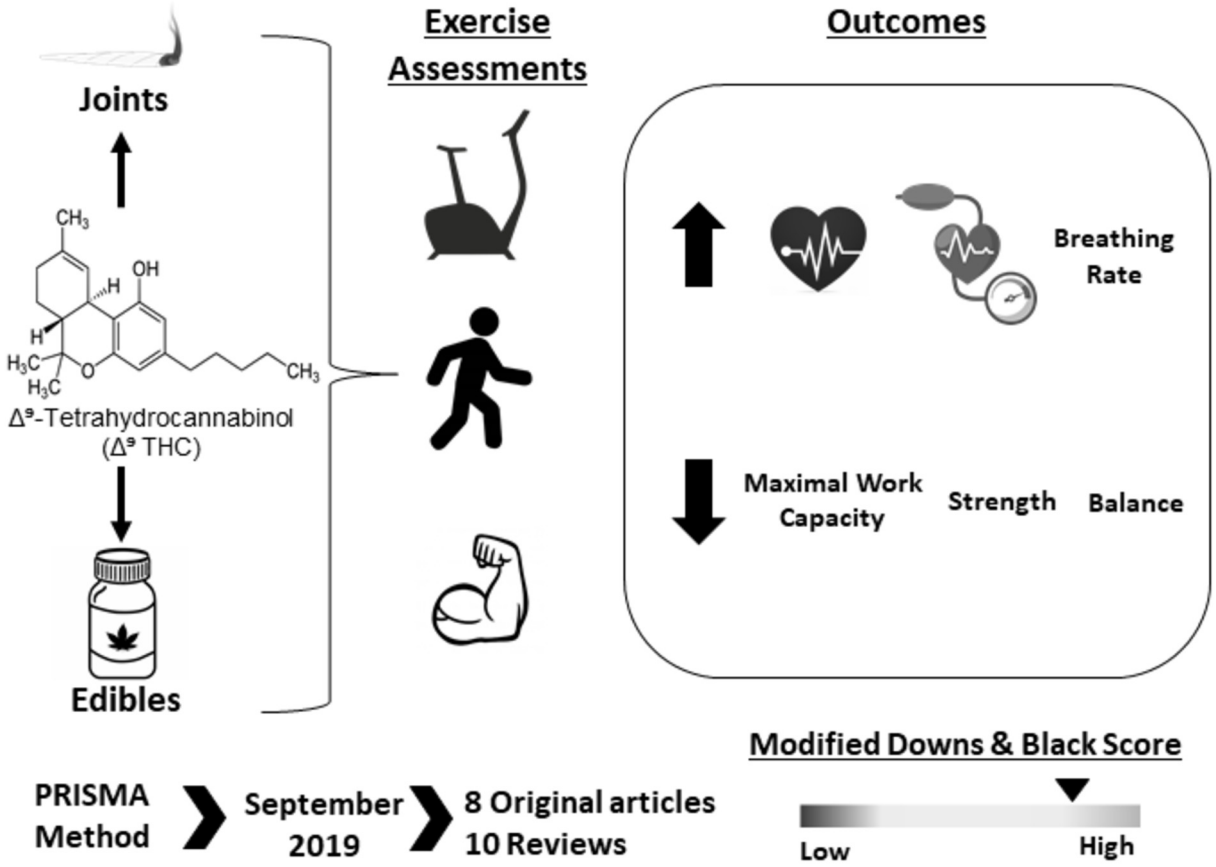

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